

Context-Aware Natural Language Processing for Interpretable Analysis of Pharmaceutical Deviation Narratives

Author: Erin Xu

Manager: Saeed Babaeizadeh

Email: saeed.babaeizadeh@takeda.com

Team: R&D Quality

Github repository:

Purpose

This document is intended for technical stakeholders to reference when reviewing the proof-of-concept implementation of a context-aware NLP enrichment and retrieval workflow for pharmaceutical deviation records. It describes the problem statement, objectives, data flow, and storage organization, source inputs, reference-data inputs, outputs produced by the pipeline, and justification of this methodology.

Problem Statement

Quality Data Navigator (QDN) provides a governed platform for structured quality data, but the narrative text associated with pharmaceutical deviations remains difficult to query and analyze systematically. These narratives contain operational context, referenced entities, recurring themes, and root-cause signals that are not fully represented in existing structured fields.

Objectives

This project develops a context-aware NLP enrichment and retrieval workflow for pharmaceutical deviation records. Rather than analyzing only the deviation title and description, the workflow constructs tiered text representations across event narratives, quality and impact assessments, structured root-cause descriptions, and action text, at both event and action-record grains. These representations support entity extraction and normalization, contextual enrichment, embeddings, topic and root-cause analysis, vector indexing, and semantic and hybrid retrieval.

Success is measured by the ability to:

1. Demonstrate that heterogeneous free-text fields can be consolidated into reproducible core, mid, and full deviation representations while preserving event and action-record grain.
2. Extract and normalize relevant entity-level information from the available narrative and descriptive fields.
3. Connect deviation records with relevant contextual and reference sources.
4. Produce enriched text suitable for embeddings and downstream NLP analysis.
5. Support both concept-based and exact-term retrieval.
6. Identify interpretable topics and root-cause signals.
7. Retain links to source fields and supporting evidence for review.
8. Produce a supplementary dataset that can be joined back to QDN data using a documented event identifier.

The following capabilities are out of scope for this project:

- Production deployment, operational support, or service-level commitments.
- Direct integration of the supplementary dataset or retrieval indexes into the production QDN application. This project produces a supplementary dataset and retrieval indexes that can be joined back to QDN data using a documented event identifier, but does not modify the production QDN application or its governed data.
- Real-time or incremental processing of newly created or updated deviation records. Production serving, continuous processing, operational monitoring, retraining, and lifecycle management require separate productionization and governance work.
- Automated quality decisions, record disposition, compliance determinations, or replacement of subject-matter-expert review.
- Changes to source-system records or existing governed QDN fields.
- Automated remediation of missing, inconsistent, or incorrect source data.
- Comprehensive extraction of every possible entity type or terminology variant.

Suggestions of these practices will be outlined in the discussion section.

Overview of Data Flow

The pipeline combines formatted TrackWise deviation records with six supporting reference datasets. Reference records are normalized into a common lookup contract and consolidated into a unified glossary. Deviation records are then compressed to one row per event, cleaned, filtered to the supported language scope, resolved against the glossary, and represented at three levels of narrative detail.

A Databricks-hosted foundation model generates a short situating context for each deviation. The deterministic reference context, generated context, and source text are combined into contextual retrieval text. BGE-M3 converts each text tier into a 1,024-dimensional dense vector. Three Databricks AI Search indexes expose the resulting core, mid, and full representations for hybrid vector and keyword retrieval.

As shown in the figure above, the pipeline resolves reference entities before generating situating context and constructing the three embedding representations.

Storage Organization

All persistent data objects used by the proof of concept are located in the development Unity Catalog.

Logical layer	Namespace	Contents
Development catalog	us_gmsgq_dev	Catalog containing the proof-of-concept data objects
Mart layer	us_gmsgq_dev.gms_us_mart	Read-only formatted deviation source
Analytics layer	us_gmsgq_dev.gms_us_alyt	Reference inputs, normalized lookups, enrichment tables, embeddings, and retrieval indexes

The mart table is treated as a read-only input. The pipeline writes derived data products to the analytics schema. These are development resources and do not represent a production deployment.

Input: Formatted TrackWise Deviation Dataset

Attribute	Value
Logical name	Formatted TrackWise deviation data
Fully qualified table	us_gmsgq_dev.gms_us_mart.tw_deviation_data_formatted_rdq

Attribute	Value
Pipeline role	Primary source of deviation identifiers, narrative fields, assessments, root-cause fields, and action text
Read/write behavior	Read only
Pipeline event identifier	<code>Event_Number</code>
Derived event identifier	<code>pr_id</code> , created by casting <code>Event_Number</code> to string
Target processing grain	One row per deviation event

Input: Source Fields Used

Field category	Source fields	Use in pipeline
Event identifier	<code>Event_Number</code>	Groups records and becomes <code>pr_id</code>
Structured clinical context	<code>Study_Protocol</code> , <code>Program_Number</code>	Additional sources for clinical-ID resolution
Structured document context	<code>Document_or_Process</code>	Additional source for controlled-document extraction
Core narrative	<code>Event_Title</code> , <code>Event_Description</code>	Concise representation of the primary deviation
Event-level context	<code>Impact_Assessment</code> , <code>Quality_Final_Assessment</code> , <code>Root_Cause_Category</code> , <code>Root_Cause_SubCategory</code>	Adds assessment and recorded root-cause information
Action-level context	<code>Action_Text</code>	Adds associated action information to the most detailed representation

Reference-Data Inputs

The pipeline uses six reference datasets to resolve entities found in deviation records.

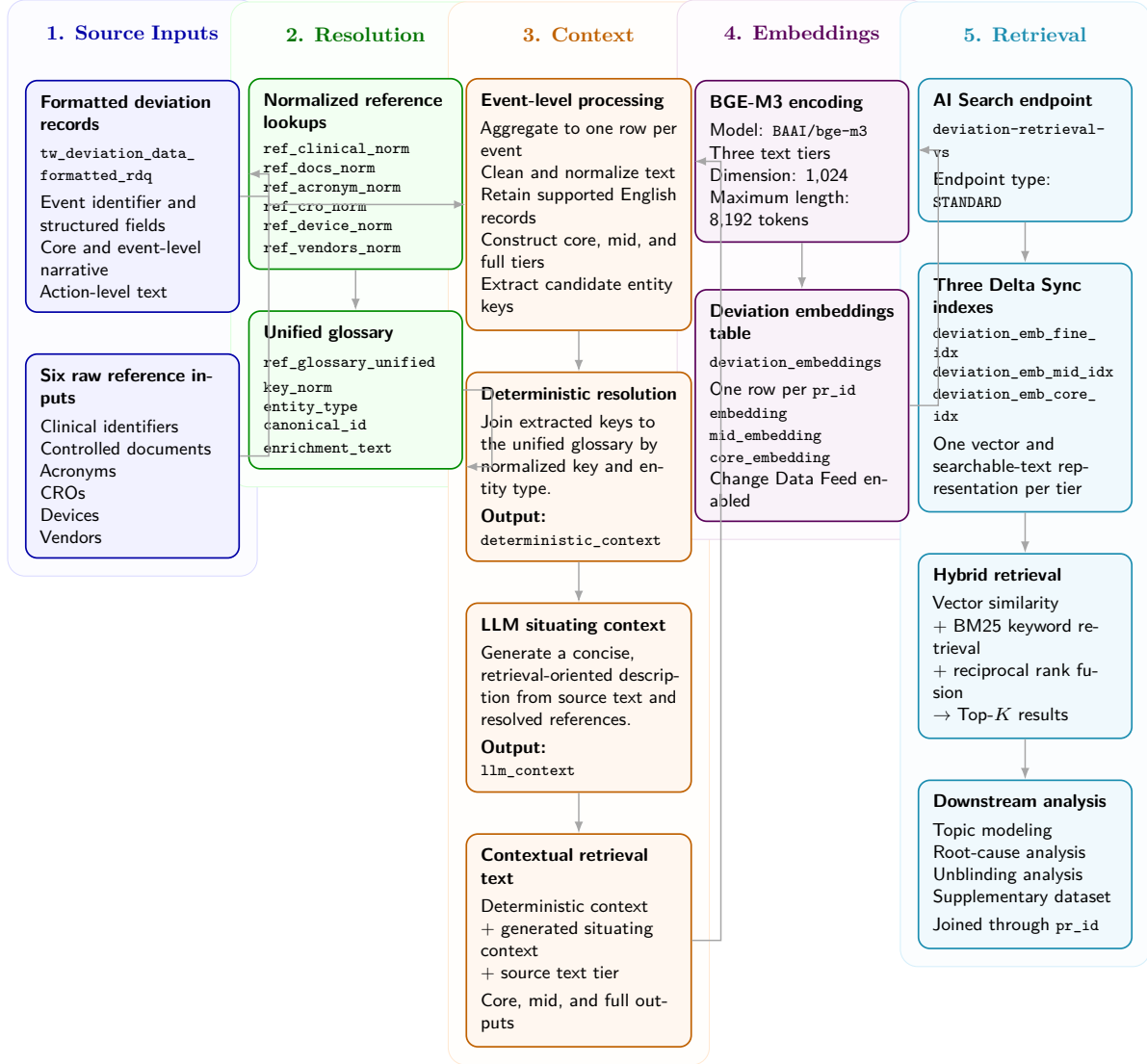


Figure 1: Principal data flow from source and reference data through contextual enrichment, embeddings, and hybrid retrieval.

Entity domain	Raw input table	Primary source fields used	Pipeline purpose
Clinical identifiers	us_gmsgq_dev.gms_us_alyt.Clinicalid_deviations	Parent_Node, Name, Protocol Number, Alternate_Name, Development_Name, Parent_Alias, Grand_Parent, Grandparent_Alias, and available contextual attributes	Resolve clinical program, compound, protocol, and study identifiers
Controlled documents	us_gmsgq_dev.gms_us_alyt.documents_number_deviations	document_number__v, previous_document_number__c, legacy_numbers__c, title__v	Resolve current, previous, and legacy controlled-document numbers
Acronyms	us_gmsgq_dev.gms_us_alyt.Acronyms_other_deviations	Acronym, Definition, Category	Resolve supported medical and industry acronyms
CROs	us_gmsgq_dev.gms_us_alyt.cro_deviations	Name, Display Name, description	Resolve known CRO names and add descriptive context
Devices	us_gmsgq_dev.gms_us_alyt.rd_device_list_deviations	Name, Alias, Device_Long_name, Device_Type, business_owner	Resolve device-name variants and add type and ownership context
Vendors	us_gmsgq_dev.gms_us_alyt.supplier_list_deviations	Supplier_Name_RD, Country_RD	Resolve known external supplier names and country context

Input: Event-Grain Reconciliation

The source table may contain multiple rows for the same `Event_Number` because `Action_Text` is handled at a row grain below the deviation event. Before downstream enrichment, the pipeline compresses the source to one row per event:

- Event-stable fields use the first available non-null value.
- Distinct non-null `Action_Text` values are collected and joined into a single text field.
- `Event_Number` is cast to a string and renamed `pr_id`.
- Duplicate action text is removed during aggregation.

This produces the event-level grain expected by the embedding table, vector indexes, and supplementary outputs.

Output: Text Tiers

The pipeline constructs three source-text representations for every retained event.

Tier	Implementation field	Included source fields	Intended role
Core	<code>source_free_text_core</code>	<code>Event_Title</code> and <code>Event_Description</code>	Concise representation of the primary deviation narrative
Mid	<code>source_free_text_mid</code>	Core fields plus <code>Impact_Assessment</code> , <code>Quality_Final_Assessment</code> , <code>Root_Cause_Category</code> , and <code>Root_Cause_SubCategory</code>	Event-level representation with assessment and root-cause context
Full	<code>source_free_text_full</code>	Mid fields plus aggregated <code>Action_Text</code>	Most detailed representation, including action-record information

The BGE-M3 and AI Search implementation also refers to the full tier as the **fine** tier. In this document, **full** describes the text representation and **fine index** refers to the corresponding implementation object.

Output: Common Lookup Contract

Each entity-specific reference builder produces a normalized table with the same required columns.

Output field	Meaning
<code>key_norm</code>	Normalized key used to match a source mention to a reference record
<code>entity_type</code>	Entity domain, such as <code>CLINICAL_ID</code> , <code>DOCUMENT</code> , <code>ACRONYM</code> , <code>CRO</code> , <code>DEVICE</code> , or <code>VENDOR</code>
<code>canonical_id</code>	Preferred identifier or display value for the resolved entity
<code>enrichment_text</code>	Human-readable reference context added to the deviation record

Additional entity-specific fields may exist during transformation, but the persistent normalized lookups expose this four-column contract.

Output: Normalized Lookup Tables

Entity type	Normalized output table	Normalization and enrichment summary
CLINICAL_ID	us_gmsgq_dev.gms_us_alyt.ref_clinical_norm	Creates keys from available clinical aliases and adds generic name, modality, therapeutic area, finance grouping, indication, target, and mechanism where available
DOCUMENT	us_gmsgq_dev.gms_us_alyt.ref_docs_norm	Normalizes current, previous, and legacy document numbers; maps the legacy MTD prefix to MTHD; adds document title
ACRONYM	us_gmsgq_dev.gms_us_alyt.ref_acronym_norm	Retains supported Medical and Industry definitions, removes short or invalid entries, excludes identifier-like values, and consolidates definitions by acronym
CRO	us_gmsgq_dev.gms_us_alyt.ref_cro_norm	Uses normalized CRO names as keys and adds name, display name, and description
DEVICE	us_gmsgq_dev.gms_us_alyt.ref_device_norm	Creates lowercase name, alias, and long-name variants and adds device type and expanded business-owner context
VENDOR	us_gmsgq_dev.gms_us_alyt.ref_vendors_norm	Uses normalized supplier names as keys and adds supplier name and country
All supported entities	us_gmsgq_dev.gms_us_alyt.ref_glossary_unified	Unions the six normalized lookup tables and retains one row per key_norm and entity_type

Entity-Specific Resolution Rules

Clinical Identifiers

Clinical identifiers are resolved from three source paths:

1. Pattern-based extraction from the combined deviation text.
2. Values in **Study_Protocol**, split on semicolons and treated as trusted clinical values after removal of null-like entries.
3. Values in **Program_Number**, split on semicolons and retained only when the leading token matches a supported clinical-ID family.

The implemented identifier families include supported TAK, MLN, SHP, HGT, CCT, DEN, and internal C-number patterns. Both canonical complete identifiers and normalized parent or compound keys may be retained for matching.

Controlled Documents

The pipeline extracts supported SOP, SPEC, MTHD, MTD, PROC, TOOL, FORM, and WI identifiers from free text and `Document_or_Process`. Current, previous, and legacy document-number fields from the reference input are normalized to a common key format.

Acronyms

Acronym extraction follows a known-set approach:

- Candidate uppercase expressions are detected in source text.
- Only candidates present in the normalized acronym reference are retained.
- One- and two-character entries are excluded.
- Values resembling supported clinical or document identifiers are excluded to avoid entity-type collisions.
- Medical and Industry definitions are retained.
- Selected British spelling variants are normalized before definition deduplication.

CROs, Devices, and Vendors

CROs, devices, and vendors are matched against known normalized names using substring matching:

- CRO names shorter than four characters are excluded.
- Device names, aliases, and long names shorter than four characters are excluded.
- Vendor names shorter than four characters are excluded.
- Device business-owner abbreviations are expanded where mappings are configured.
- Matching remains dependent on the coverage and quality of each reference table.

These methods favor interpretable, reference-grounded matches. They do not constitute comprehensive named-entity recognition for all possible organizations, devices, suppliers, or terminology variants.

Entity Resolution Output

Entity keys extracted from each deviation are transformed into event-and-entity records containing:

- `pr_id`
- `key_norm`
- `entity_type`

Duplicate combinations of `pr_id`, `key_norm`, and `entity_type` are removed. The extracted keys are then left-joined to the unified glossary by `key_norm` and `entity_type`.

Successful matches contribute their `enrichment_text` to an event-level field named `deterministic_context`. Multiple resolved references for the same event are deduplicated and concatenated into one context block.

Unmatched extracted keys remain available in the diagnostic workflow but do not contribute reference enrichment to `deterministic_context`.

Contextualized Embedding Input

Persistent Enrichment Table

Attribute	Value
Fully qualified table	<code>us_gmsgq_dev.gms_us_alyt.deviation_embed_input</code>
Grain	One row per retained <code>pr_id</code>
Primary purpose	Store source text, deterministic reference context, generated situating context, and final contextual retrieval text
Write behavior	Overwrite through a staging table
Temporary staging tables	<code>deviation_embed_input_staging</code> and <code>deviation_embed_input_stg</code>
Downstream consumers	Embedding generation and retrieval workflow

Output Fields

Field group	Fields	Description
Event identifier	<code>pr_id</code>	String representation of <code>Event_Number</code>
Source text	<code>source_free_text_full</code> , <code>source_free_text_mid</code> , <code>source_free_text_core</code>	Cleaned source text at three levels of detail
Deterministic enrichment	<code>deterministic_context</code>	Resolved and concatenated reference information
Generated enrichment	<code>llm_context</code>	Short model-generated paragraph situating the deviation using source and resolved reference context
Contextual retrieval text	<code>contextual_retrieval_text_full</code> , <code>contextual_retrieval_text_mid</code> , <code>contextual_retrieval_text_core</code>	Final text passed to embeddings and synchronized for keyword retrieval

Each contextual retrieval field is assembled in this order: Resolved deterministic references + LLM-generated situating context + Source free text for the selected tier.

The original source-text tiers remain stored separately so that generated or resolved context does not replace source evidence.

Generated Situating Context

The pipeline calls a Databricks-hosted foundation model through `ai_query`. The configured model identifier is:

```
databricks-gpt-5-4-mini
```

For each non-empty event, the prompt supplies:

- `deterministic_context`
- `source_free_text_full`
- Instructions to produce a short retrieval-oriented description of the entities involved and the subject of the deviation

The configured generation parameters are:

Parameter	Configured value
Maximum generated tokens	300
Temperature	0.0

The generated result is stored as `llm_context` and reused across the full, mid, and core contextual retrieval representations. It is generated context, not source evidence. Reviewers should use the retained source fields and deterministic references when validating its contents.

Embedding Output

Deviation Embeddings Table

Attribute	Value
Fully qualified table	<code>us_gmsgq_dev.gms_us_alyt.deviation_embeddings</code>
Grain	One row per <code>pr_id</code>
Primary key	<code>pr_id</code>
Change Data Feed	Enabled
Embedding model	BAAI/bge-m3
Embedding dimension	1,024
Configured maximum sequence length	8,192 tokens
Configured batch size	64
Number of text tiers	Three
Write behavior	Overwrite for the first write chunk; append for subsequent chunks

Embedding Fields

Text tier	Contextual text input	Vector output
Full/fine	<code>contextual_retrieval_text_full</code>	<code>embedding</code>
Mid	<code>contextual_retrieval_text_mid</code>	<code>mid_embedding</code>
Core	<code>contextual_retrieval_text_core</code>	<code>core_embedding</code>

The three text lists are combined into one encoding workload, sorted approximately by token length to reduce unnecessary padding, encoded, restored to their original order, and split back into the three vector columns.

The embedding table also retains:

- `source_free_text_full`
- `source_free_text_mid`
- `source_free_text_core`
- `deterministic_context`
- `llm_context`
- `contextual_retrieval_text_full`
- `contextual_retrieval_text_mid`
- `contextual_retrieval_text_core`

Retaining these fields preserves the relationship between each vector and the text from which the vector was generated.

Retrieval Outputs

AI Search Endpoint

Attribute	Value
Endpoint name	<code>deviation-retrieval-vs</code>
Endpoint type	STANDARD
Environment	Development
Authentication	Notebook credentials detected by the AI Search client

Delta Sync Indexes

Tier	Index	Vector field	Searchable text field
Full/fine	<code>us_gmsgq_dev.gms_us_alyt.deviation_emb_fine_idx</code>	<code>embedding</code>	<code>contextual_retrieval_text_full</code>
Mid	<code>us_gmsgq_dev.gms_us_alyt.deviation_emb_mid_idx</code>	<code>mid_embedding</code>	<code>contextual_retrieval_text_mid</code>
Core	<code>us_gmsgq_dev.gms_us_alyt.deviation_emb_core_idx</code>	<code>core_embedding</code>	<code>contextual_retrieval_text_core</code>

Each index:

- Uses `deviation_embeddings` as its source table.
- Uses `pr_id` as the primary key.
- Uses a precomputed 1,024-dimensional BGE-M3 vector.
- Uses a triggered Delta Sync pipeline.
- Synchronizes the tier-specific contextual retrieval text for keyword retrieval.
- Does not ask Databricks AI Search to recompute the stored embeddings.

At query time, the retrieval workflow supplies:

- A BGE-M3 query vector for semantic similarity.
- The original query text for keyword matching.
- HYBRID as the query type.

Databricks AI Search returns the combined ranking from vector and keyword retrieval. The notebook returns the top configured results for the selected tier.

Input-to-Output Lineage

Input	Principal transformation	Persistent output	Downstream use
<code>Clinicalid_deviations</code>	Alias parsing, clinical-ID normalization, contextual-field assembly	<code>ref_clinical_norm</code>	Unified glossary and deterministic enrichment
<code>documents_number_deviations</code>	Current, previous, and legacy document normalization	<code>ref_docs_norm</code>	Unified glossary and deterministic enrichment
<code>Acronyms_other_deviations</code>	Filtering, normalization, definition consolidation	<code>ref_acronym_norm</code>	Acronym extraction, glossary enrichment, and selected abbreviation expansion
<code>cro_deviations</code>	CRO-name filtering and normalization	<code>ref_cro_norm</code>	Known-set CRO extraction and enrichment
<code>rd_device_list_deviations</code>	Name, alias, and long-name normalization	<code>ref_device_norm</code>	Known-set device extraction and enrichment
<code>supplier_list_deviations</code>	Supplier-name normalization	<code>ref_vendors_norm</code>	Known-set vendor extraction and enrichment

Input	Principal transformation	Persistent output	Downstream use
Six normalized references	Schema alignment, union, and deduplication	ref_glossary_unified	Single reference join for all supported entity types
tw_deviation_data_formatted_rdq	Event aggregation, text cleaning, language filtering, text-tier construction, and entity extraction	deviation_embed_input	Context generation and embedding
deviation_embed_input	BGE-M3 dense encoding of three contextual text tiers	deviation_embeddings	Vector and hybrid retrieval
deviation_embeddings	Triggered Delta Sync indexing	Three AI Search indexes	Core, mid, and full/fine retrieval

Temporary and Diagnostic Outputs

The following objects or results support safe execution and validation but are not canonical downstream data products.

Artifact	Purpose	Persistence
deviation_embed_input_staging	Materializes the event-level embedding input before replacing the Temporary; dropped after the write canonical table	
deviation_embed_input_stg	Materializes model-generated context before replacing the Temporary; dropped after the write canonical table	
Entity extraction coverage	Counts events with at least one extracted key by entity type Notebook diagnostic	
Reference join coverage	Compares extracted keys with keys matched to the unified glossary Notebook diagnostic	
Unmatched-key report	Lists frequently extracted keys without a reference match Notebook diagnostic	
LLM-context coverage	Counts populated and empty llm_context values	Notebook diagnostic
Embedding integrity checks	Checks row counts, distinct pr_id values, vector dimensions, Notebook diagnostic and context coverage	
Index-status checks	Reports index state and indexed-row counts Notebook diagnostic	

These diagnostics should be retained in the handoff workflow because they provide the principal evidence that the transformation completed as expected.

Canonical Outputs

The canonical outputs of this workflow are:

1. Six normalized entity-specific lookup tables.
2. One unified reference glossary.
3. One event-level contextualized embedding-input table.
4. One event-level table containing source text, context, and three embedding vectors.
5. Three tier-specific Delta Sync retrieval indexes.

The canonical output tables and indexes are:

1. `us_gmsgq_dev.gms_us_alyt.ref_clinical_norm`
2. `us_gmsgq_dev.gms_us_alyt.ref_docs_norm`
3. `us_gmsgq_dev.gms_us_alyt.ref_acronym_norm`
4. `us_gmsgq_dev.gms_us_alyt.ref_cro_norm`
5. `us_gmsgq_dev.gms_us_alyt.ref_device_norm`
6. `us_gmsgq_dev.gms_us_alyt.ref_vendors_norm`
7. `us_gmsgq_dev.gms_us_alyt.ref_glossary_unified`
8. `us_gmsgq_dev.gms_us_alyt.deviation_embed_input`
9. `us_gmsgq_dev.gms_us_alyt.deviation_embeddings`
10. `us_gmsgq_dev.gms_us_alyt.deviation_emb_fine_idx`
11. `us_gmsgq_dev.gms_us_alyt.deviation_emb_mid_idx`
12. `us_gmsgq_dev.gms_us_alyt.deviation_emb_core_idx`

Technical Dependencies

The notebook installs or uses the following principal dependencies:

Dependency	Role
PySpark	Source aggregation, reference transformations, joins, and Delta table writes
langdetect	Reproducible language detection for the supported English-language scope
FlagEmbedding	BGE-M3 model loading and dense vector generation
transformers	Model and tokenizer dependency
databricks-ai-search	Endpoint and Delta Sync index management and querying

Dependency	Role
NumPy	Vector-order restoration and local array handling
pandas	Driver-side transfer for embedding generation
Databricks <code>ai_query</code>	Server-side generation of the situating context
Hugging Face model access	Initial retrieval of the <code>BAAI/bge-m3</code> model artifacts
GPU-capable driver	FP16 BGE-M3 embedding generation in the current implementation

The package installation cell should not be treated as a version-locked environment specification because most dependencies are not pinned to exact versions. For reproducible handoff, record the tested Databricks Runtime, Python version, package versions, AI Search SDK version, BGE-M3 model revision, and foundation-model endpoint configuration.

Text Preparation and Filtering

Before text construction, the pipeline:

1. Applies Unicode normalization.
2. Standardizes Unicode hyphens and non-breaking spaces.
3. Folds accented Latin characters for matching.
4. Normalizes repeated spaces and excessive blank lines.
5. Preserves letter case because acronym and CRO extraction depend on case-sensitive source information.
6. Applies language detection to the combined `Event_Title` and `Event_Description`.
7. Retains records classified as English, including supported code-only records.
8. Applies an additional guard against remaining CJK characters in the title and description.
9. Excludes records whose resulting full text is empty.

Language detection is seeded to make repeated runs reproducible.

The English-language filter is based on the core title and description, not independently on every source field. This behavior should be retained in the documented acceptance criteria and reviewed if multilingual coverage is later required.